

Lab Assignment 6

Course: AI - Assisted Coding

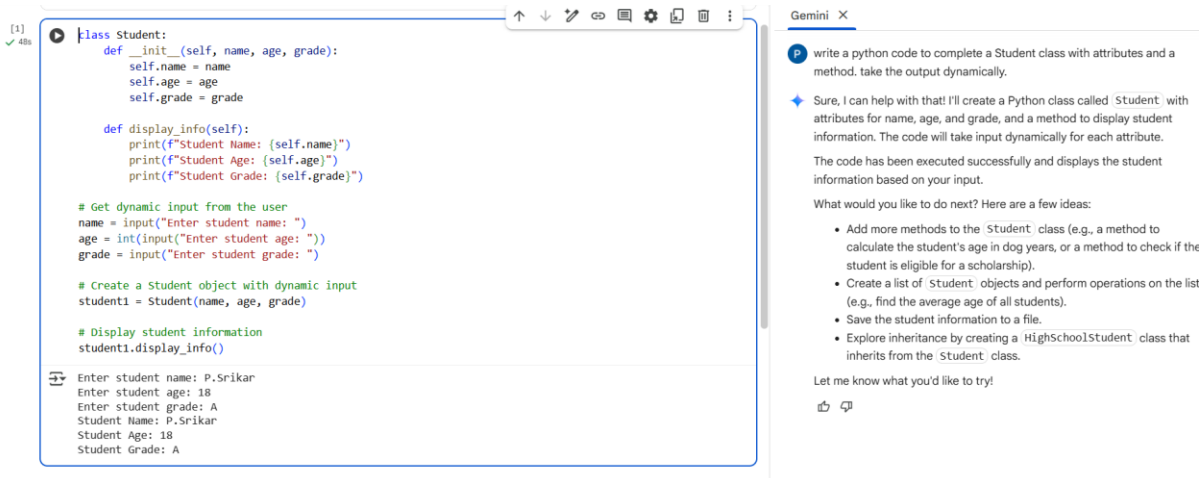
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Task-1:

Prompt: write a python code to complete a Student class with attributes and a method. take the output dynamically.

Code&Output:



```
[1] ✓ 48s
class Student:
    def __init__(self, name, age, grade):
        self.name = name
        self.age = age
        self.grade = grade

    def display_info(self):
        print(f"Student Name: {self.name}")
        print(f"Student Age: {self.age}")
        print(f"Student Grade: {self.grade}")

# Get dynamic input from the user
name = input("Enter student name: ")
age = int(input("Enter student age: "))
grade = input("Enter student grade: ")

# Create a Student object with dynamic input
student1 = Student(name, age, grade)

# Display student information
student1.display_info()
```

Enter student name: P.Srikar
Enter student age: 18
Enter student grade: A
Student Name: P.Srikar
Student Age: 18
Student Grade: A

Gemini X

write a python code to complete a Student class with attributes and a method. take the output dynamically.

Sure, I can help with that! I'll create a Python class called 'Student' with attributes for name, age, and grade, and a method to display student information. The code will take input dynamically for each attribute. The code has been executed successfully and displays the student information based on your input.

What would you like to do next? Here are a few ideas:

- Add more methods to the 'Student' class (e.g., a method to calculate the student's age in dog years, or a method to check if the student is eligible for a scholarship).
- Create a list of 'Student' objects and perform operations on the list (e.g., find the average age of all students).
- Save the student information to a file.
- Explore inheritance by creating a 'HighSchoolStudent' class that inherits from the 'Student' class.

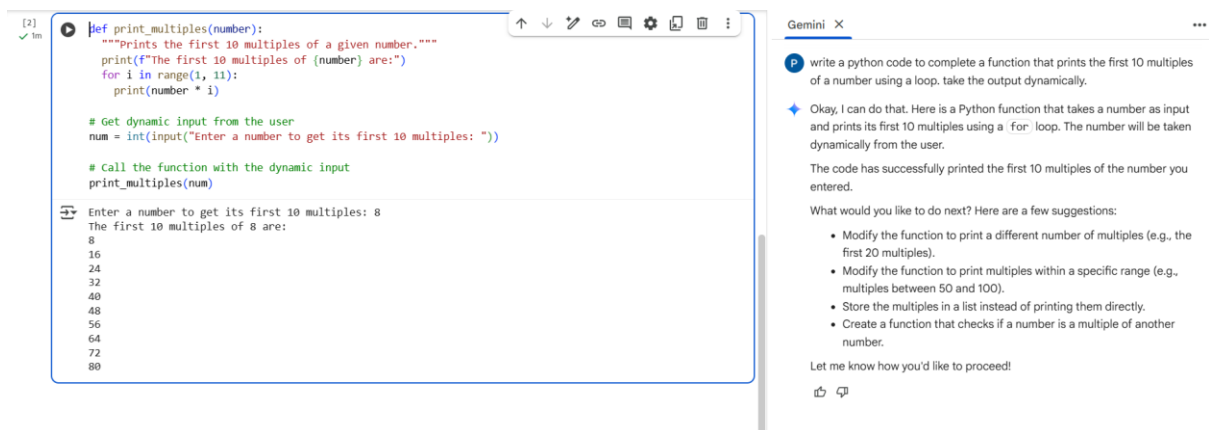
Let me know what you'd like to try!

Observation: The task successfully demonstrated how to create a Python class with attributes and a method that takes user input to populate those attributes. This is a fundamental concept in object-oriented programming for creating custom data structures and behaviors.

Task-2:

Prompt: write a python code to complete a function that prints the first 10 multiples of a number using a loop. take the output dynamically.

Code&Output:



```
[2] ✓ 1m def print_multiples(number):  
    """Prints the first 10 multiples of a given number."""  
    print(f"The first 10 multiples of {number} are:")  
    for i in range(1, 11):  
        print(number * i)  
  
    # Get dynamic input from the user  
    num = int(input("Enter a number to get its first 10 multiples: "))  
  
    # Call the function with the dynamic input  
    print_multiples(num)  
  
Enter a number to get its first 10 multiples: 8  
The first 10 multiples of 8 are:  
8  
16  
24  
32  
40  
48  
56  
64  
72  
80
```

Gemini X

write a python code to complete a function that prints the first 10 multiples of a number using a loop. take the output dynamically.

Okay, I can do that. Here is a Python function that takes a number as input and prints its first 10 multiples using a `for` loop. The number will be taken dynamically from the user.

The code has successfully printed the first 10 multiples of the number you entered.

What would you like to do next? Here are a few suggestions:

- Modify the function to print a different number of multiples (e.g., the first 20 multiples).
- Modify the function to print multiples within a specific range (e.g., multiples between 50 and 100).
- Store the multiples in a list instead of printing them directly.
- Create a function that checks if a number is a multiple of another number.

Let me know how you'd like to proceed!

Observation: This task successfully demonstrated how to define a Python function that takes user input and uses a for loop to iterate and perform a calculation (multiplication) a specific number of times. It also included basic error handling for invalid input.

Task-3:

Prompt: write a python code to write nested if-elif-else conditionals to classify age groups. take the output dynamically.

Code&Output:

The screenshot shows a code editor on the left and a Gemini chat window on the right. The code editor contains a Python script for age classification. The script prompts the user to enter an age, checks for negative values, and then uses nested if-elif-else statements to classify the age into groups: Child (0-12), Teenager (13-19), Adult (20-64), and Senior (65+). The user has entered '18', and the output is 'Teenager'. The Gemini chat window on the right shows a prompt asking for Python code to write nested if-elif-else conditionals. The Gemini response provides the code snippet and offers suggestions for modifications, such as adding more age groups or using a single if-elif-else block.

```
[3] ✓ 7s
# Get dynamic input from the user
age = int(input("Enter age: "))

# Classify age groups using nested if-elif-else conditionals
if age < 0:
    print("Age cannot be negative.")
else:
    if age <= 12:
        print("Child")
    elif age <= 19:
        print("Teenager")
    elif age <= 64:
        print("Adult")
    else:
        print("Senior")

Enter age: 18
Teenager
```

Gemini X

write a python code to write nested if-elif-else conditionals to classify age groups. take the output dynamically.

Alright, I can help you with that. I'll write a Python code snippet that uses nested if-elif-else statements to classify an age input into different groups. The age will be taken dynamically from the user.

The code has successfully classified the age you entered into an age group.

What would you like to do next? Here are a few ideas:

- Modify the age ranges for the different groups.
- Add more age groups (e.g., "Toddler" or "Young Adult").
- Use a different conditional structure, like a single if-elif-else block instead of a nested one, to achieve the same result.
- Take the age input as a string and handle potential errors if the user enters non-numeric input.

Let me know what you'd like to explore!

Observation: This task successfully demonstrates how to use a while loop to repeatedly take user input and apply nested if-elif-else conditions to classify each input. It also includes a mechanism for the user to exit the input loop.

Task-4:

Prompt: Write a python code to Generate a sum_to_n() function to calculate sum of first n numbers. take the output dynamically.

Code&Output:

The screenshot shows a code editor on the left and a Gemini chat window on the right. The code editor contains a Python script that defines a sum_to_n() function, which calculates the sum of the first n numbers using a for loop. The script prompts the user to enter a number n, calls the function, and prints the result. The user has entered '8', and the output is 'The sum of the first 8 numbers is: 36'. The Gemini chat window on the right shows a prompt asking for Python code to generate a sum_to_n() function. The Gemini response provides the code snippet and offers suggestions for modifications, such as exploring alternative ways to calculate the sum or handling non-integer input.

```
[4] ✓ 29s
def sum_to_n(n):
    """calculates the sum of the first n numbers."""
    total_sum = 0
    for i in range(1, n + 1):
        total_sum += i
    return total_sum

# Get dynamic input from the user
num = int(input("Enter a number (n) to calculate the sum of the first n numbers: "))

# Calculate and print the sum
result = sum_to_n(num)
print(f"The sum of the first {num} numbers is: {result}")

Enter a number (n) to calculate the sum of the first n numbers: 8
The sum of the first 8 numbers is: 36
```

Gemini X

Write a python code to Generate a sum_to_n() function to calculate sum of first n numbers. take the output dynamically.

Certainly! I can write a Python function called sum_to_n() that calculates the sum of the first 'n' numbers. The value of 'n' will be taken as dynamic input from the user.

The code has successfully calculated and displayed the sum of the first 'n' numbers based on your input.

What would you like to do next? Here are a few ideas:

- Explore alternative ways to calculate the sum of the first 'n' numbers, such as using a mathematical formula ($n * (n + 1) / 2$).
- Modify the function to calculate the sum of numbers within a specific range.
- Calculate other mathematical series using a similar approach (e.g., sum of squares, sum of cubes).
- Handle potential errors if the user enters non-integer input.

Let me know what you'd like to do!

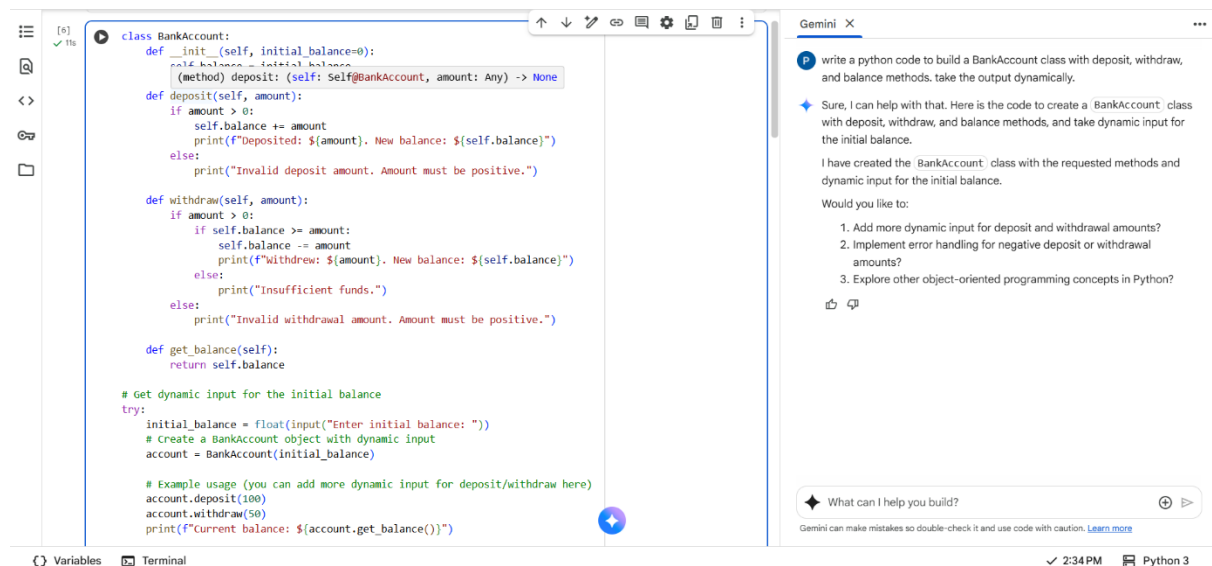
Observation: This task successfully demonstrates how to define a Python function that takes user input, validates it,

and uses a for loop to iterate and accumulate a sum. This is a common pattern for performing calculations over a range of numbers.

Task-5:

Prompt: write a python code to build a BankAccount class with deposite,withdraw,and balance methods.take the output dynamically.

Code&Output:



The image shows a code editor with a Python script for a `BankAccount` class. The code includes methods for `deposit`, `withdraw`, and `get_balance`, along with a dynamic input section for the initial balance and example usage.

```
class BankAccount:
    def __init__(self, initial_balance=0):
        self.balance = initial_balance
    (method) deposit: (self: Self@BankAccount, amount: Any) -> None
    def deposit(self, amount):
        if amount > 0:
            self.balance += amount
            print(f"Deposited: ${amount}. New balance: ${self.balance}")
        else:
            print("Invalid deposit amount. Amount must be positive.")

    def withdraw(self, amount):
        if amount > 0:
            if self.balance >= amount:
                self.balance -= amount
                print(f"Withdrew: ${amount}. New balance: ${self.balance}")
            else:
                print("Insufficient funds.")
        else:
            print("Invalid withdrawal amount. Amount must be positive.")

    def get_balance(self):
        return self.balance

# Get dynamic input for the initial balance
try:
    initial_balance = float(input("Enter initial balance: "))
    # Create a BankAccount object with dynamic input
    account = BankAccount(initial_balance)

    # Example usage (you can add more dynamic input for deposit/withdraw here)
    account.deposit(100)
    account.withdraw(50)
    print(f"Current balance: ${account.get_balance()}")
```

The right side of the image shows the Gemini chat interface. The prompt is: "write a python code to build a BankAccount class with deposit, withdraw, and balance methods. take the output dynamically." The response is: "Sure, I can help with that. Here is the code to create a BankAccount class with deposit, withdraw, and balance methods, and take dynamic input for the initial balance. I have created the BankAccount class with the requested methods and dynamic input for the initial balance. Would you like to: 1. Add more dynamic input for deposit and withdrawal amounts? 2. Implement error handling for negative deposit or withdrawal amounts? 3. Explore other object-oriented programming concepts in Python?"

The screenshot displays a code editor with a Python script for a `BankAccount` class. The class includes methods for `deposit`, `withdraw`, and `get_balance`. The script also handles dynamic input for the initial balance and includes error handling for invalid input. The output of the script is shown in the terminal at the bottom.

```
print(f"Deposited: ${amount}. New balance: ${self.balance}")
else:
    print("Invalid deposit amount. Amount must be positive.")

def withdraw(self, amount):
    if amount > 0:
        if self.balance >= amount:
            self.balance -= amount
            print(f"Withdrew: ${amount}. New balance: ${self.balance}")
        else:
            print("Insufficient funds.")
    else:
        print("Invalid withdrawal amount. Amount must be positive.")

def get_balance(self):
    return self.balance

# Get dynamic input for the initial balance
try:
    initial_balance = float(input("Enter initial balance: "))
    # Create a BankAccount object with dynamic input
    account = BankAccount(initial_balance)

    # Example usage (you can add more dynamic input for deposit/withdraw here)
    account.deposit(100)
    account.withdraw(50)
    print(f"Current balance: ${account.get_balance()}")

except ValueError:
    print("Invalid input. Please enter a valid number for the initial balance.")
```

Terminal output:

```
Enter initial balance: 10000
Deposited: $100. New balance: $10100.0
Withdrew: $50. New balance: $10050.0
Current balance: $10050.0
```

The Gemini chat interface on the right shows a prompt to write a Python class for a bank account. The response provides the code and suggests further enhancements like dynamic input, error handling, and exploring object-oriented programming concepts.

Observation: This task successfully demonstrates the creation of a Python class with multiple methods (`deposit`, `withdraw`, `get_balance`) that encapsulate data (`balance`) and behavior. It also incorporates a loop for continuous user interaction and basic error handling for invalid input, simulating a simple banking application.